Komodo Mlipir Algorithm to Solve Generator Scheduling Problems

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2022 2nd International Conference on Electronic and Electrical Engineering and Intelligent System (ICE3IS) | 978-1-6654-6541-0/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICE3IS56585.2022.10010294

***Abstract*—Economic dispatch (ED) is one of the main problems in optimizing the electrical power system. The main goal of ED is to create an optimal schedule for all power plants to generate the minimum cost while complying with the ramp rate limits. Previously, numerous approaches have been made to solve ED, using mathematically and artificially based methods. This study proposes the use of an artificial method based on the state-of-the-art optimization algorithm known as the Komodo Mlipir Algorithm (KMA). The obtained result shows that the proposed method can outperform previous and other optimization methods. The test is conducted to calculate the ED from 42 power plants in the Jawa-Madura-Bali (Jamali) electricity system, Indonesia, which consists of thermal and hydro-power plants. The results highlight that KMA can be considered an alternative solution to ED.**

Keywords—Artificial-based, Economic Dispatch, Jamali, KMA, optimal schedule , ramp rate.

I. INTRODUCTION

One of the fundamental optimation problems in power system operation is the economic dispatch (ED). ED is used to calculate the total generation cost of the generator unit [1]. The optimal ED decides the production level for each generator to fulfill the needs of the scheduled load demands [2]–[5]. The main objective of ED is to obtain the minimum generating unit costs while satisfying the limitations [6]. The main expense in the calculation of ED is the cost of fuel, which typically accounts for between 60%-80% of the entire production cost. Therefore it is essential to optimize fuel consumption through an efficient dispatch. Numerous approaches to resolving generator scheduling issues have been investigated, including the mathematically based (MM) and artificially based (AM) methods.

MM is specifically used for differentiable functions [7], [8], and the LaGrange technique is observed by [7], [9] to find the solution using iterative steps such as the Newton, gradient, and lambda iteration methods. Another method, such as the direct method (DM), is developed by Zein et al. [8], based on mathematical derivation to obtain formulas with direct solutions. This method can be applied to a large system and provide a shorter computation time. However, this method is limited to the second-order objective function. MM can obtain high accuracy, and the solution certainty falls to a global minimum. This condition is often used to validate other methods. The optimal point is obtained by deriving the LaGrange function so that the MM is unable to complete the optimization of the non-differentiable function [10].

The AM method can be applied to various types of functions, both differentiable and non-differentiable functions. However, multiple factors, including control variables, can affect the accuracy. Several AM methods have been published [11]–[17]. There are two concepts developed in AM to achieve the optimal point, the point-to-point and area reduction. To reach the solution point, the first method process the optimization through a point-to-point approach. These methods include neural network (NN), particle swarm optimization (PSO), firefly algorithms (FA), and genetic algorithms (GA). PSO and point-to-point-based algorithms have the disadvantage that they are often stuck at the local minimum for higher-order functions [12]. On the other hand, the optimal solution obtained in the area reduction method is done by reducing the feasible area to its minimal value [15]–[18]. The humpback whale optimization algorithm (WOA), The Technique of Narrow Down Area (ToNDA), the Multi Dimension of Coarse to Fine Search (MDCFS) method, and the large to small area technique (LSAT) are AM methods based on area reduction.

The WOA algorithm performs better than PSO and GA, but WOA has a weakness if each iteration has similarities and non-linear inequalities take longer [15]. The ToNDA method has high accuracy in solving ED problems and can compete with the QPSO, SSA, IPSO-TVAC, HCRO-DE, SHDE, HGA, AND FAPSO-VDE methods [16]. The MDCFS method can compete with the PSO method in solving ED problems; even the optimal solution is very close to the DM optimal solution [17]. Meanwhile, the LSAT method, which involves restricted operating area constraints (POZ), can outperform the Genetic Algorithm (GA) method. Furthermore, this method is able to compete with the results from Lambda Iteration (LI) and Dynamic programming (DP) [18]. The involvement of unit commitment (UC) in the ED shows a significant advantage where the results can accurately classify the generators in bearing base load, load fluctuation, and peak load.

Previous studies showed that population-based metaheuristic optimization algorithms such as GA, PSO, FA, and WOA were inferior compared to other methods. The aforementioned methods were based on the natural phenomena or animal behaviors in creating a searching

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strategy to create balance in exploitation-exploration. There was a possibility of getting stuck on searches on local optima. A novel algorithm named Komodo Mlipir Algorithm (KMA) was proposed by Suyanto et al. to tackle the limitation of other metaheuristic algorithms [19]. The KMA could avoid the local optima trap and tackle the plateaus area, making it outperform other state-of-the-art metaheuristic algorithms by achieving global optima for 17 out of 23 benchmark functions. However, it has never been tested using real-world data. Therefore, this study applies KMA to solve the optimization problem in generator scheduling and compares it to one of the state-of-the-art metaheuristic algorithms named Slime Mould Algorithm (SMA). SMA used adaptive weight to calculate the propagation wave feedback of slime mould. It has excellent ability to explore and exploitation propensity [20].

This paper is organized as follows. The second section discusses the proposed use of KMA for formulating the generator scheduling. The third section presents the experimental result using data from Indonesian Power Plants. The fourth involved the discussion and the conclusion of this study.

II. Problem Formulation

There are several restrictions, including the parameters of each generator, the objective functions, and ramp rate. Those parameters must be fulfilled by the KMA in order to solve the economic dispatch issue.

A. Economic Dispatch

The main goal of economic dispatch is to reduce the overall cost of generating enough power to fulfill the demands. This process is done on an hourly basis by choosing which of the plants to operate to meet the power demand. The optimization must satisfy an equality and inequality constraints. This study analyzes the ED for 42 power plants in Jawa-Madura-Bali (Jamali) electricity system, Indonesia, which consist of thermal and hydro-power plants.

B. Objective Functions

Each power plant generator has their own cost fucntion, which can be simply expressed as a quadratic function. The expression of the standard mathematical techniques is shown in (1) and (2) [21].

$$TC = \sum_{i=1}^{J} F_i(P_i) \tag{1}$$

$$F_i(P_i) = a_i {P_i}^3 + b_i {P_i}^2 + c_i P_i \tag{2}$$

Here, the TC represents the total generation cost while F_i and P_i represent the cost function and the electrical output of generators, respectively. The cost coefficient of generators is expressed with a_i , b_i , and c_i. The number of all generators is represented with set J.

C. Constraints

The overall generation cost of the power plants should be minimized and equal with the whole system demand in Jamali electricity system. The objective function should comply with (3).

$$\min F_i(P_i) \tag{3}$$

Since the network loss is not taken into account in this study, the equality constraint is expressed with (4).

$$\sum_{i=1}^{J} P_i = P_D \tag{4}$$

Here, the P_D shows the total demand for the system. Moreover, the generated output from the system must be obtained between the minimum and maximum values, see (5).

$$P_{i,min} \le P_i \le P_{i,max} \tag{5}$$

This study involves the ramp-rate as the constraint. Therefore, the power limitation in (5) is changed hourly as shown in (6) and (7).

$$P_{i,\min} = \max\{P_{i,\min}, P_i(t-1) - RD_i\} \tag{6}$$

$$P_{i,\max} = \max\{P_{i,\max}, P_i(t-1) - RU_i\} \tag{7}$$

Here, the $P_{i,min}$ and $P_{i,max}$ represent the i-th generator's minimum and maximum power limits, respectively. The i-th generator's ramp down and ramp up are denoted by RD and RU, respectively.

D. Power Plant Data

The data used in this study is gathered from 42 power plants, consist of electric steam and hydroelectric power plants, from Jamali electricity system, Indonesia. The characteristic of each power plant is presented, including the ramp rate limit. The detail of the data is shown in Table I. The daily load demand used in this study is taken from the existing Jamali electricity system, which customized to fit with the selected 42 power plants. The data is shown in Fig.1.

TABLE I. Fuel cost coefficient of 42 power plants from Jamali electricity system, Indonesia

PP	a	b	c	P_{min}	P_{max}	RU	RD
	(IDR)	(IDR/MW)	(IDR/MW^2)	(MW)		(MW/h)	
1	1.06	-796.5	500770.41	250	371.5	300	300
2	1.06	-796.5	500770.41	250	371.5	300	300
3	1.06	-796.5	500770.41	250	371.5	300	300
4	1.06	-796.5	500770.41	250	371.5	300	300
5	0.53	-580.731	548169.063	408	575	300	300
6	0.53	-580.731	548169.063	408	575	300	300
7	0.53	-580.731	548169.063	408	575	300	300
8	0	0	10000	87.5	174.6	720	720
9	0	0	10000	87.5	174.6	720	720
10	0	0	10000	87.5	174.6	720	720
11	0	0	10000	87.5	174.6	720	720
12	-29.86	2116.1058	578259.403	40	80	60	60
13	-29.86	2116.1058	578259.403	40	80	60	60
14	-1.88	253.0934	437774.976	75	170	120	120
15	-1.88	253.0934	437774.976	75	170	120	120
16	1.661	-1475.578	674957.404	220	480	300	300
17	1.661	-1475.578	674957.404	220	480	300	300
18	1.661	-1475.578	674957.404	220	480	300	300
19	967.659	-26243.28	837352.902	5	16	120	120
20	967.659	-26243.28	837352.902	5	16	120	120
21	0	0	15000	60	118.5	7200	7200
22	0	0	15000	60	118.5	7200	7200
23	0	0	15000	60	118.5	7200	7200
24	0	0	15000	60	118.5	7200	7200
25	0	0	15000	60	118.5	7200	7200
26	0	0	15000	60	118.5	7200	7200
27	0	0	15000	60	118.5	7200	7200
28	0	0	15000	60	118.5	7200	7200
29	0.67	-803.4	664191.82	400	661	180	180
30	0.67	-803.4	664191.82	400	661	180	180
31	0.67	-803.4	664191.82	400	661	180	180
32	0.67	-803.4	664191.82	400	661	180	180
33	-5.98	61.9058	620176.752	120	140	240	240
34	2.429	-2101.891	934134.238	200	459	240	240
35	2.429	-2101.891	934134.238	200	459	240	240
36	0.536	-727.78	828790.263	345	640	1200	1200
37	1.071	-1494.774	1142590.01	180	274	300	300
38	4.045	-3723.275	1659160.91	200	420	540	540
39	4.045	-3723.275	1659160.91	200	420	540	540
40	13.045	-5341.694	1068419.16	180	214	420	420
41	1.36	-963.92	541408.16	250	370	240	240
42	1.36	-963.92	541408.16	250	370	240	240

*PP: Power Plant; RU: Ramp Up; RD: Ramp Down

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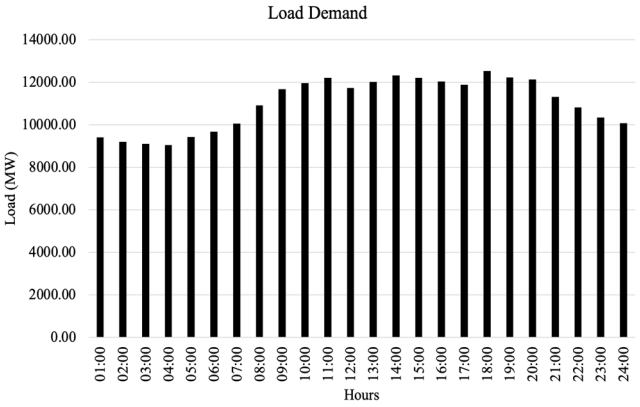


Fig. 1. Daily load demand

E. Komodo Mlipir Algorithm

Komodo Mlipir Algorithm (KMA) is a state-of-the-art metaheuristic algorithm proposed by Suyanto et al. [19], which was inspired by the behavior of Komodo dragons and the Javanese term *mlipir* (a Javanese phrase that means to walk along the side of the road safely). This algorithm adopts the hunting and reproduction movement of Komodo dragons. KMA splits the solution candidate into three groups following the Komodo dragon individuals: big males, female, and small males. The KMA algorithm is started by the movement of big males known as the high-exploitation low-exploration (HILE), then followed by the female, which attempts to find a solution by mating with the best big male (highest-quality), or through parthenogenesis (exploration). The last is the low-quality males diversifying the potential solution by utilizing the *mlipir* movement.

The Komodo population is split into three groups using (8) and (9). The portion of the splitting Komodo groups is denoted as p using the interval (0-1). This portion is then used to split n number of Komodo into q big males, one female, and s small males. After that, the movement of each big male (k_i) produces a new position (k_i') expressed by (10) and (11).

$$q = \lfloor (p-1)n \rfloor \tag{8}$$

$$s = n - q \tag{9}$$

$$k_i' = k_i + \sum_{j=1}^{q} wij \text{ , where } j \neq i \tag{10}$$

$$w_{ij} = \begin{cases} r_1(k_j - k_i) \text{ , if } f(k_j) < f(k_i) \text{ or } r_2 < 0.5 \\ r_1(k_i - k_j) \quad \text{, otherwise} \end{cases} \tag{11}$$

The Komodo movement using (10) is done to keep the q highest-quality big males survive in the next generation. The female mating the big male winner using (12) and (13) or doing parthenogenesis using (14).

$$k_{il}' = r_l . k_{il} + (1 - r_l) . k_{jl} \tag{12}$$

$$k_{jl}' = r_l . k_{jl} + (1 - r_l) . k_{il} \tag{13}$$

$$k_{ij}' = k_{ij} + (2r - 1)\alpha \left| ub_j - lb_j \right| \tag{14}$$

$$\{k_{i1}, k_{i2}, \dots, k_{im}\} \rightarrow \{k_{i1}', k_{i2}', \dots, k_{im}'\} \tag{15}$$

The k_{il} and k_{jl} are the l-th dimension of the i-th and the j-th winning big male and the female Komodos, respectively. The mating process produce two offsprings, which are the k_{il}' and the k_{jl}' while a random number interval [0,1] is shown as r_k. In parthenogenesis, the process is done by adding a small value to each female dimension, which shown in (15).

The small males movement is defined by (16) and (17) then keeping their position in order to survive in the next generation. The last process is updating the population size n using (17). Detailed process of KMA can be found in [19]

$$w_{ij} = \begin{cases} \sum_{l=1}^{m} r_1(k_{jl} - k_{il}), \text{if } r_2 < d \\ 0, \text{otherwise} \end{cases} \tag{16}$$

$$k_i' = k_i + \sum_{j=1}^{q} w_{ij} \text{ , where } j \neq i \tag{17}$$

$$n' = \begin{cases} n - a, \text{if } \delta f_1 > 0 \text{ and } \delta f_2 > 0 \\ n + a, \text{if } \delta f_1 = 0 \text{ and } \delta f_2 = 0 \end{cases} \tag{18}$$

III. Simulation Result

ED's main goal is to find the minimum production cost. KMA is used to calculate the optimal cost by following the objective fuction for each power plant. The process is shown in Fig. 2. ED is calculated for the data from 42 power plants in the Jamali electricity system. The daily data is simulated to produce hourly generated data from each power plant to fulfill the daily load following the constraints shown in Table I. The output of this process is shown in Fig. 3.

The KMA used in this study has three adjustable parameters they are population size (n), *mlipir* rate (d), and portion (p). This study used 15 million evaluations, with the initial 5,000 evaluations using a population of 5 individuals, a half portion of big males (p_l), maximum *mlipir* rate ($d_1 = \frac{(m-1)}{m}$), and the rest of the evaluation used a self-adaptive population size scheme from 100 to 1,000 individuals.

The result in Fig. 3 shows that the KMA produces a higher load calculation in the first six hours than the load demand. The highest ratio for the first six hours is 1.16 (03.00). However, during the rest hours, KMA can fulfill the load demands, especially during peak hours. This condition occurs because the working principle of KMA is to complete ED simultaneously for 24 hours at a time, not hour by hour. Thus, the high loads that have a dominant influence on costs will be resolved first. However, this condition can be overcome by performing longer iterations, but it will cost a longer time. Judging from the main objective of ED is to minimize costs, then KMA can be used as an alternative solution to ED problems.

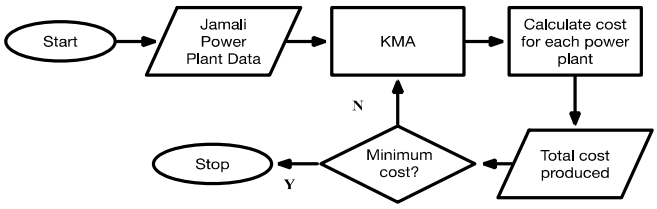


Fig. 2. General process of solving generator scheduling problem using KMA

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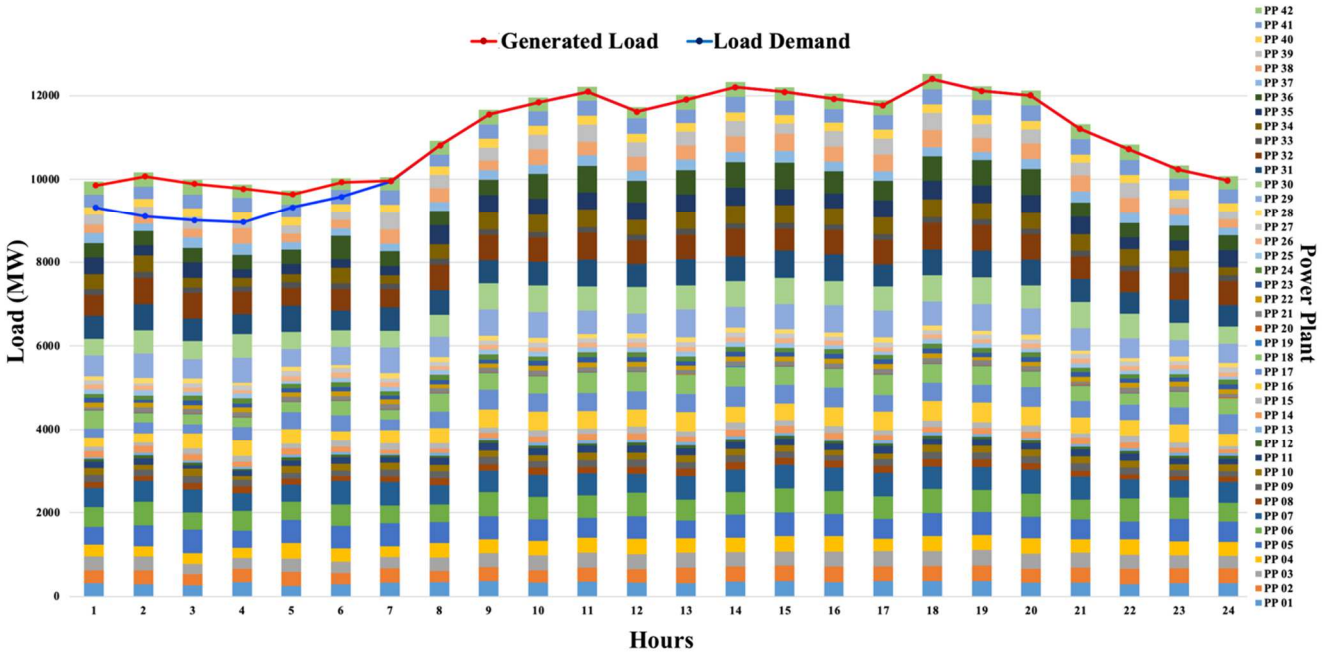


Fig. 3. Hourly generated data compared to the load demand from 42 power plants

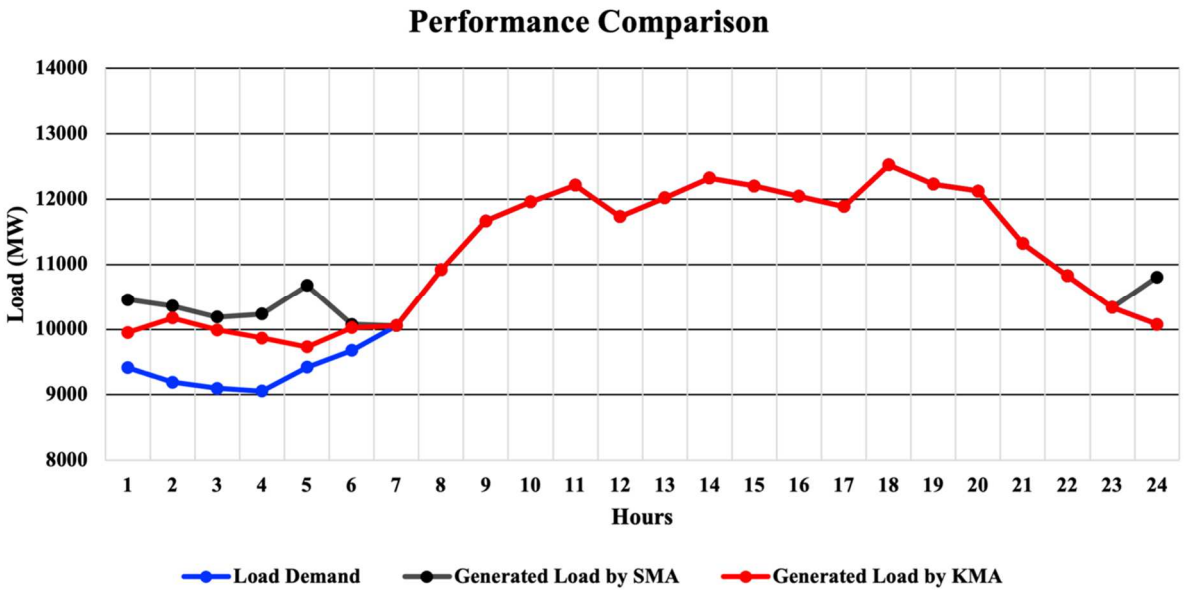


Fig. 4. Comparison of KMA and SMA in ED

TABLE II. COST COMPARISON WITH OTHER METHODS

Method	No. of PP	Cost (IDR/hr.)	Cost (USD/hr.)
SMA	42	4,573,286,328.00	304,885.76
KMA	42	4,479,537,419.61	298,635.83

The result of KMA in ED is compared with another state-of-the-art optimization algorithm to strengthen the performance of KMA, which is the Slime Mould Algorithm (SMA). The same scheme and constraints are tested with the SMA, and the comparison result is shown in Fig. 4. The graph shows that in the first six hours, the SMA produces loads with a ratio ranging from 1.04 (06.00) to 1.13 (02.00, 04.00, and 05.00). On the other hand, KMA produces better performance with a ratio ranging from 1.03 (05.00) to 1.11 (02.00). Furthermore, the SMA has been unable to fulfill the load demand in the last hour with the ratios of 1.07.

Further comparison is made by observing the cost. It is shown in Table II that KMA can produce at a lower cost compared with SMA. This shows that KMA has better accuracy with no violation of the constraints. The use of KMA to solve the ED that involving more generators and tested using bigger daily load with an average 11,013.46 MW can achieve the average cost of USD298,635.83. The use of KMA can reduce the cost significantly. Furthermore, this study shows that KMA outperformed the SMA by producing a 2.05% lower cost.

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IV. CONCLUSION

This study presents a new approach to calculating ED problems based on the KMA. The ED is calculated for 42 power plants from Jamali electric system in Indonesia. The experimental result indicates KMA provides better performance than the comparison methods and can reduce costs by 2.05% using SMA. Thus KMA can be considered an alternative solution to solve ED.

ACKNOWLEDGMENT

This research is funded by PT. PLN through the Memorandum of Understanding No. 006.Pj/HKM.02.01/ C30000000/2022 and No. 034/SAM4/RC-DBE/2022 and Letter of Assignment of PT PLN Persero No. 1152/LIT.00.02/ C30000000/2022.

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